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SUBJECT CODE NO: - RNP-8002

FACULTY OF SCIENCE AND TECHNOLOGY

S.E (Civil Engg.) (NEP) Sem - III

Examination November / December 2025 (To be held in February-2026)

Fluid Mechanics

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.no 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve any 10 following questions

20

1. Which fluid obeys Newton's law of viscosity?
A) Plastic fluid B) Ideal fluid
C) Newtonian fluid D) Pseudo plastic fluid
2. Reynolds number is the ratio of
A) Inertia force to gravity force
B) Viscous force to inertia force
C) Inertia force to viscous force
D) Pressure force to viscous force
3. Total Energy Line (TEL) always lies
A) Below HGL
B) Coinciding with HGL
C) Above HGL
D) None of these
4. At datum Manning's equation is commonly used for
A) Pipe flow B) Laminar flow
C) Open channel flow D) Compressible flow
5. Cavitation in pumps occurs due to
A) High pressure
B) Low pressure below vapour pressure
C) High velocity only
D) Blocked delivery pipe

6. Reaction turbines work under
 - A) Atmospheric pressure only
 - B) Partial pressure
 - C) Vacuum
 - D) Pressure energy and kinetic energy
7. Define metacentre and metacentric height.
8. Define steady and unsteady flow.
9. State Pascal's law.
10. Define siphon. State its application.
11. What is pipe network analysis?
12. Define cavitation.
13. Define slip of a reciprocating pump.

SECTION B

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| Q2 | a) Define and derive hydrostatic law | 05 |
| | b) Write a note on Equilibrium of floating body and meta centric height | 05 |
| Q3 | a) Explain the properties of fluids | 05 |
| | b) A circular plate 3 m diameter is immersed in water in such a way that its greatest and least depth below the free surface are 4 m and 1.5 m respectively. Determine the total pressure on one face of the plate and position of the centre of pressure. | 05 |
| Q4 | a) Explain major and minor losses in pipes | 05 |
| | b) Find the head lost due to friction in a pipe of diameter 300 mm and length 50 m, through which water is flowing at a velocity of 3 m/s using (i) Darcy formula, (ii) Chezy's formula for which $C = 60$. Take ν for water = 0.01 stoke. | 05 |
| Q5 | a) Explain the concept of equivalent pipes and branching of pipes. | 05 |
| | b) Explain the working of a siphon with a neat sketch. | 05 |
| Q6 | a) Write short notes on Draft tube. | 05 |
| | b) Explain multistage centrifugal pump | 05 |
| Q7 | a) A Pelton wheel is to be designed for the following specifications: Shaft power = 11,772 kW; Head = 380 metres; Speed = 750 r.p.m.; Overall efficiency = 86%; Jet diameter is not to exceed one-sixth of the wheel diameter. Determine: (i) The wheel diameter, (iii) Diameter of the jet. (ii) The number of jets required, and Take $K_v = 0.985$ and $K_{s,} = 0.45$ | 05 |
| | b) Write note on different types of efficiencies in turbine. | 05 |